

The Double-Ended Helix: A Three-Dimensional Geometric Tool for the Vanishing Points of Dirichlet L -Functions

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Abstract

We present a three-dimensional geometric tool: a double-ended chiral helix carrier on which a phasor fiber identifies the vanishing points — the crossings where the phasor accumulation cancels — that coincide with the zeros of the Riemann zeta function and Dirichlet L -functions. The carrier is a growing-radius helix with integers placed at uniform arclength spacing $\Delta = \pi/3$; a fiber rides it and accumulates phasors. The fiber attached to a Dirichlet character χ is the partial-sum sequence $\sum_{n < N} \chi(n) n^{-s}$; its limit is the Dirichlet L -function $L(s, \chi)$ on $\operatorname{Re} s > 1$, and, after Abel summation, on the whole half-plane $\operatorname{Re} s > 0$, the critical line included. The angular winding of the carrier is a completely multiplicative character built from prime angles by the fundamental theorem of arithmetic, with no logarithm; the logarithm enters only in the external bridge wind $n \leftrightarrow n^{it}$ that identifies the geometric winding with the analytic phasor. We record the geometric explicit formula (the logarithmic derivative of the fiber is the winding-weighted von Mangoldt prime field, and each simple zero is a simple pole with an explicit residue), the reality of the completed object on the critical line (read through a projection that rescales the helix's $\pi/3$ gauge to a unit coordinate), the conjugacy of the two chiralities of the double-ended carrier, and a determinant identity: at a conjugate crossing the two chiral eigenphases $z = e^{-iy \log n}$ and $\bar{z}$ assemble the transverse block $\operatorname{diag}(z, \bar{z})$, whose determinant is $z\bar{z} = |z|^2 = 1$. The coincidence of the fiber's vanishing points with the actual zeros has been verified numerically for the leading zeros of ζ and ten Dirichlet L -functions. Every statement below is formalized in Lean 4 with Mathlib and checked by the kernel; the axiom footprint of each named theorem is {propxet, Classical.choice, Quot.sound}, with no `sorry` and no additional axioms.

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1 Objects and notation

Let χ be a Dirichlet character modulo q and $L(s, \chi)$ the associated L -function; ζ is the Riemann zeta function and $\Lambda(s) = \text{completedRiemannZeta}(s)$ its completed form, normalized so that $\Lambda(1-s) = \Lambda(s)$. The *critical line* is the set $s = \frac{1}{2} + iy$, $y \in \mathbb{R}$. We write $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ for the unit circle (`Mathlib.Circle`) and $\bar{}$ for complex conjugation (`starRingEnd  $\mathbb{C}$`). For an arithmetic weight χ the *phasor* attached to the integer $n \geq 1$ on the vertical line $s = \sigma + iy$ is $\chi(n) n^{-s}$.

The classical analytic theory of ζ and Dirichlet L -functions — the Euler product, the functional equation, and the von Mangoldt explicit formula [1, 2, 3, 4, 5, 7] — enters here through the corresponding `Mathlib` lemmas.

All named results carry the identifier of the corresponding Lean declaration, e.g. `frobenius_conjugate_det_one`; Section 11 records the verification.

2 The $\pi/3$ helix carrier

2.1 The space curve and its arclength

Definition 2.1 (Carrier curve; `Geometry.helix`, `helixVel`, `speed`). *For pitch p and radial rate r , the carrier is the growing-radius helix*

$$\gamma(k) = (r k \cos 2\pi k, r k \sin 2\pi k, p k), \quad k \in \mathbb{R}.$$

Its velocity is $\gamma'(k) = (r \cos 2\pi k - 2\pi r k \sin 2\pi k, r \sin 2\pi k + 2\pi r k \cos 2\pi k, p)$, and the squared speed is

$$\|\gamma'(k)\|^2 = p^2 + r^2 + (2\pi r k)^2.$$

The climber is $k(y) = e^y/p$, so the height is $z = p k(y) = e^y$ and the cylindrical radius is $R(y) = r k(y) = (r/p) e^y$.

Definition 2.2 (`Arclength`; `Geometry.arclength`, `arclengthClosed`). *The arclength of γ from 0 to k is $S(k; p, r) = \int_0^k \sqrt{p^2 + r^2 + (2\pi r t)^2} dt$.*

Theorem 2.3 (Closed-form arclength; `arclength_closed_form`, `arclength_r_zero`). For $r > 0$,

$$S(k; p, r) = \frac{k}{2} \sqrt{p^2 + r^2 + 4\pi^2 r^2 k^2} + \frac{p^2 + r^2}{4\pi r} \operatorname{arsinh}\left(\frac{2\pi r k}{\sqrt{p^2 + r^2}}\right),$$

and $S(k; p, 0) = p k$ for $p \geq 0$.

Proof sketch. Differentiate the stated antiderivative and check it equals the integrand $\sqrt{p^2 + r^2 + 4\pi^2 r^2 k^2}$ pointwise, then apply the fundamental theorem of calculus on $[0, k]$; the constant-pitch case is $\int_0^k p dt$. $\square$

2.2 Uniform integer placement and the area law

Definition 2.4 (Spacing, index, integer angle; `Geometry.Delta`, `Nindex`, `spinAngle`). The fixed spacing is $\Delta = \pi/3$. The continuous geometric index is $N(y) = S(k(y); p, r)/\Delta$, so $N(y) = (3/\pi) S(k(y); p, r)$ (`Nindex_eq`). The integer angular coordinate is $s_n = n \cdot (\pi/3)$.

Theorem 2.5 (Uniform $\pi/3$ spacing; `carrier_spacing`). For every $n \in \mathbb{N}$, $s_{n+1} - s_n = \pi/3$.

Theorem 2.6 (Emergent radius; `HelixLogFree.norm_helixPt`). Let $\text{helixPt}(n) = \sqrt{n} \cdot (\text{wind } n)$ (Definition 3.1). The distance of the n -th site from the axis is

$$\|\text{helixPt}(n)\| = \sqrt{n},$$

so the enclosed area $\pi \|\text{helixPt}(n)\|^2 = \pi n$ is linear in n . The winding contributes unit modulus (Theorem 3.2); the factor $\sqrt{n}$ is read off the placement, not posited. The uniform spacing $\Delta = \pi/3$ enters separately, through the index $N(y) = S(k(y); p, r)/\Delta$ of Definition 2.4 (`helixPt` is the unit-gauge carrier point).

2.3 The mod-6 bucket structure

Because $6 \cdot (\pi/3) = 2\pi$, the integer angle s_n is 6-periodic on the circle.

Theorem 2.7 (Bucket periodicity and counts; `spin_phasor_period6`, `bucket_count`). $\exp(i s_{n+6}) = \exp(i s_n)$ for all n , and for $a < 6$ exactly N of the integers in $[0, 6N)$ satisfy $n \equiv a \pmod{6}$. Thus the integers collapse onto six angular directions, each receiving an equal share.

Remark 2.8 (Sixth roots of unity; Eisenstein integers). The integer-angle phasor $\exp(i s_n) = \exp(in\pi/3)$ takes values in the sixth roots of unity $\mu_6 = \{e^{ik\pi/3} : 0 \leq k < 6\}$, cycling through them as n runs mod 6; the primitive sixth root $\zeta = e^{i\pi/3}$ ($\zeta^2 = \zeta - 1$, the cyclotomic relation Φ_6) generates μ_6 . The ring $\mathbb{Z}[\zeta]$ is the ring of Eisenstein integers [8] — a hexagonal (triangular) lattice whose unit group is exactly μ_6 . So the $\pi/3$ gauge is the Eisenstein symmetry, and the six buckets of Theorem 2.7 are its units.

3 The log-free winding and the bridge

3.1 The fundamental-theorem-of-arithmetic winding

Definition 3.1 (Winding angle and winding; `HelixLogFree.windAngle`, `wind`, `helixPt`). Fix a prime-angle assignment θ (in Lean, a function $\theta : \mathbb{N} \rightarrow \mathbb{R}$; only its values on primes enter). The

winding angle is the completely additive extension read off the prime factorization,

$$\Theta(n) = \sum_{p^e \parallel n} e \theta(p) = \text{n.factorization.sum}(\lambda p e. e \cdot \theta(p)),$$

and the winding is $\text{wind}(n) = \exp(i \Theta(n)) \in \mathbb{T}$. The carrier point is $\text{helixPt}(n) = \sqrt{n} \cdot \text{wind}(n)$. No logarithm occurs in Θ .

Theorem 3.2 (Multiplicative character; `windAngle_mul`, `wind_mul`, `wind_one`). For $m, n \neq 0$, $\Theta(mn) = \Theta(m) + \Theta(n)$ and $\text{wind}(mn) = \text{wind}(m) \text{wind}(n)$, with $\text{wind}(1) = 1$. Thus $\text{wind} : \mathbb{N} \rightarrow \mathbb{T}$ is a completely multiplicative character [6].

Proof sketch. Θ is a sum over `Nat.factorization`; additivity is `Nat.factorization_mul` (the fundamental theorem of arithmetic). Exponentiating turns additivity into multiplicativity on $\mathbb{T}$ via `Circle.exp_add`. $\square$

3.2 The bridge $\text{wind } n \leftrightarrow n^{it}$

The single place a logarithm enters is the external identification of the geometric winding with the analytic phasor.

Theorem 3.3 (Logarithm is FTA-additive; `real_log_eq_factorization_sum`). For $n \neq 0$, $\log n = \sum_{p^e \parallel n} e \log p$.

Theorem 3.4 (The bridge; `windAngle_glog`, `wind_glog_eq_cpow`). With the prime-angle assignment $\theta(p) = \gamma \log p$, one has $\Theta(n) = \gamma \log n$ and hence

$$\text{wind}(n) = n^{i\gamma} \quad (n \geq 1).$$

Proof sketch. Substitute Theorem 3.3 into Θ to get $\Theta(n) = \gamma \log n$; then $\exp(i\gamma \log n) = n^{i\gamma}$ by the principal-branch definition of $n^{i\gamma}$ for the positive real base n . $\square$

4 The phasor fiber and its accumulation

4.1 Phasor decomposition

Theorem 4.1 (Vertical-line phasor and magnitude; `cpow_vertical_line_phasor`, `norm_cpow_vertical_line`). For a positive real base x and $s = \sigma + iy$,

$$x^{-s} = x^{-\sigma} \exp(-(y \log x) i), \quad |x^{-s}| = x^{-\sigma}.$$

In particular, on the critical line $n^{-(1/2+iy)} = n^{-1/2} \exp(-(y \log n) i)$ with magnitude $n^{-1/2}$ for every y (`cpow_critical_line`, `norm_cpow_critical_line`).

Definition 4.2 (The spin; `LFunctionPhasor.spin`). The spin of the integer n at ordinate y is the unit-modulus factor $\text{spin}(y, n) = \exp(-(y \log n) i)$.

Theorem 4.3 (The fiber rides the carrier; `fiber_rides_carrier`). Write $\text{wind}_{-t \log}$ for the winding of Definition 3.1 with the assignment $\theta(p) = -t \log p$, so $\text{wind}_{-t \log}(n) = n^{-it}$ by the bridge (Theorem 3.4). For $n \geq 1$, the carrier magnitude $n^{-1/2}$ — the reciprocal of the emergent radius $\sqrt{n}$ — times this winding is the Dirichlet phasor:

$$n^{-1/2} \cdot \text{wind}_{-t \log}(n) = n^{-(1/2+it)}.$$

4.2 Accumulation to L

Definition 4.4 (Finite phasor carrier; `DirichletPhasorCarrier.finiteCarrier`, `phasorTerm`). $G_{\chi,N}(s) = \sum_{n < N} \chi(n) n^{-s}$, the partial sums of the carrier-riding phasors.

Theorem 4.5 (Accumulation on $\operatorname{Re} s > 1$; `fiber.accumulates.to.L`, `finiteCarrier.tendsto.LFunction`). For every Dirichlet character χ and $\operatorname{Re} s > 1$, $G_{\chi,N}(s) \rightarrow L(s, \chi)$ as $N \rightarrow \infty$.

5 Strip extension: accumulation onto the critical line

The accumulation converges beyond the region of absolute convergence, down to $\operatorname{Re} s > 0$.

Theorem 5.1 (Dirichlet strip extension; `dirichlet_strip_tendsto.LFunction`). For a Dirichlet character $\chi \neq 1$ modulo q and $\operatorname{Re} s > 0$,

$$\sum_{n < N} \chi(n) n^{-s} \xrightarrow{N \rightarrow \infty} L(s, \chi).$$

Theorem 5.2 (Eta strip extension; `eta_strip_tendsto`). For $\operatorname{Re} s > 0$ and $s \neq 1$,

$$\sum_{n < N} (-1)^{n+1} n^{-s} \xrightarrow{N \rightarrow \infty} (1 - 2^{1-s}) \zeta(s).$$

Proof sketch. Bucket cancellation gives a uniform bound on the character sums $\sum_{n < N} \chi(n)$; Abel summation by parts [6, 3] then yields convergence of $\sum \chi(n) n^{-s}$ on $\operatorname{Re} s > 0$, and the limit agrees with $L(s, \chi)$ on $\operatorname{Re} s > 1$ and hence everywhere on the connected strip by the identity theorem. For the alternating weight the factor $1 - 2^{1-s}$ is the even-term correction relating η and ζ . $\square$

Theorem 5.3 (Continuous reach to the line; `continuous_model.zeta`). For every $y \in \mathbb{R}$, at $s = \frac{1}{2} + iy$,

$$\sum_{n < N} (-1)^{n+1} n^{-s} \rightarrow (1 - 2^{1-s}) \zeta(\tfrac{1}{2} + iy),$$

and the limit is 0 if and only if $\zeta(\frac{1}{2} + iy) = 0$.

Proof sketch. Theorem 5.2 at $s = \frac{1}{2} + iy$ (using $\operatorname{Re} s = \frac{1}{2} > 0$ and $s \neq 1$). On the line the factor $1 - 2^{1-s}$ is nonzero (`correction_factor_ne_zero.on.critical.line`), so the product vanishes exactly when $\zeta(\frac{1}{2} + iy)$ does (`etaTrivial.eq.zero.iff`). $\square$

6 The geometric explicit formula

The logarithmic derivative of the fiber is the prime-power accumulation weighted by the winding.

Theorem 6.1 (Prime field; `explicit_formula_prime.field`, `dirichlet.logDeriv.eq.tsum`). For $\operatorname{Re} s > 1$,

$$-\frac{L'(s, \chi)}{L(s, \chi)} = \sum_n \chi(n) \Lambda(n) n^{-s},$$

where Λ is the von Mangoldt function. The zeros of $L(\cdot, \chi)$ are the poles of the meromorphic continuation of this object.

Definition 6.2 (Residue harmonic; `Residue.residueHarmonic`). For $x > 0$ and $\gamma \in \mathbb{R}$, write $\rho = \frac{1}{2} + i\gamma$ and set $\operatorname{residueHarmonic}(x, \gamma) = -x^\rho / \rho$.

Theorem 6.3 (Zeros located as poles; `zeta_zero_located_as_pole`, `L_zero_located_as_pole`). Let $\rho = \frac{1}{2} + i\gamma$ be a zero of $L(\cdot, \chi)$ at which $L'(\rho, \chi) \neq 0$ (a simple zero). Then ρ is a simple pole of the explicit-formula kernel $-(L'/L)(s)x^s/s$, with

$$\lim_{s \rightarrow \rho} (s - \rho) \left(-\frac{L'(s, \chi)}{L(s, \chi)} \cdot \frac{x^s}{s} \right) = \text{residueHarmonic}(x, \gamma) = -\frac{x^\rho}{\rho}.$$

The same statement holds verbatim for ζ .

Proof sketch. On the punctured neighborhood of ρ in $\{s \neq 1\}$, L is analytic with $L(\rho) = 0$ and $L'(\rho) \neq 0$, so $-(L'/L)$ has a simple pole at ρ with residue 1; multiplying by the analytic factor x^s/s scales the residue to x^ρ/ρ and the leading sign gives $-x^\rho/\rho$. $\square$

7 Projection and reality on the critical line

The carrier is three-dimensional; the readout projects it to a one-dimensional coordinate. We first record that coordinate and the change of scale it carries (Section 7.1), then the reality of the completed object that the readout returns (Section 7.2).

7.1 The readout coordinate and its scaling

The carrier is built in its native gauge: the unit is $\pi/3$, and a full angular turn is six such units, $6 \cdot (\pi/3) = 2\pi$ (`spin_phasor_period6`, Theorem 2.7). The readout that sends a three-dimensional harmonic value $t \in \mathbb{R}$ down to a one-dimensional coordinate composes a Cayley map [9] with a rescaling to the standard unit interval (`HarmonicProjection`):

$$\text{toCircleAngle}(t) = 2 \arctan t \in (-\pi, \pi), \quad \text{toLine}(\theta) = \frac{\theta + \pi}{2\pi} \in (0, 1),$$

and $\text{projection} = \text{toLine} \circ \text{toCircleAngle}$. The second map carries the change of scale: it sends the full turn 2π to length 1, so the six $\pi/3$ buckets of the gauge become six equal sub-intervals of the readout. Under this rescaling the coordinate is rigid — the Cayley map is odd (`toCircleAngle_odd`), the normalization is affine hence midpoint-preserving (`toLine_midpoint`), and the principal-arc ends $\theta = \pm\pi$ map to the two ends of the interval (`toLine_neg_pi`, `toLine_pi`) — so the centre of the gauge is carried to the centre of the interval.

Theorem 7.1 (Coordinate centre; `HarmonicProjection.projection_midline`). *projection(0) is the midpoint of the readout interval: the centre of the six gauge buckets, the angle $\theta = 0$, maps to the average of the two endpoint images (`toLine_midpoint_of_endpoints`).*

The inverse conversion is the same scaling read backwards: a readout coordinate u is the angle $\theta = 2\pi u - \pi$, the gauge reading $6u - 3$ in $\pi/3$ units, and the interval's centre is the gauge centre $\theta = 0$. This is a property of the readout coordinate and its scaling. The value the readout returns on the completed object is real — it lands on the real axis (Section 7.2).

7.2 Reality on the line

The supporting analytic facts are the conjugation (Schwarz reflection [9]) and functional symmetries of Λ .

Theorem 7.2 (Conjugation symmetry; `HelixCollapse.completedRiemannZeta_conj`). *For all $s \in \mathbb{C}$, $\overline{\Lambda(\bar{s})} = \Lambda(s)$.*

Proof sketch. On $\operatorname{Re} s > 1$ this is the real Dirichlet coefficients of $\Lambda(\overline{\Lambda(\bar{s})}) = \Lambda(s)$ termwise, using $\overline{\pi^{-s/2}} = \pi^{-\bar{s}/2}$ and $\overline{\Gamma(\bar{s}/2)} = \Gamma(s/2)$; the identity extends to $\mathbb{C}$ by analytic continuation of Λ_0 (the pole-removed completion) via the identity theorem. $\square$

Theorem 7.3 (Reality on the line; `completedRiemannZeta_critical_line_im_zero`). *For every $t \in \mathbb{R}$, $\operatorname{Im} \Lambda(\frac{1}{2} + it) = 0$.*

Proof sketch. On the line $\overline{\frac{1}{2} + it} = 1 - (\frac{1}{2} + it)$, so Theorem 7.2 together with $\Lambda(1-s) = \Lambda(s)$ (`completedRiemannZeta_one_sub`) gives $\Lambda(\frac{1}{2} + it) = \overline{\Lambda(\frac{1}{2} + it)}$, i.e. a real value. $\square$

Theorem 7.4 (Faithful projection for ζ ; `faithful_projection_zeta`). *For every $\gamma \in \mathbb{R}$,*

$$\operatorname{Im} \Lambda(\frac{1}{2} + i\gamma) = 0 \quad \text{and} \quad F_\eta(\gamma) = 0 \iff \zeta(\frac{1}{2} + i\gamma) = 0,$$

where $F_\eta(\gamma) = (1 - 2^{1-s})\zeta(s)$ at $s = \frac{1}{2} + i\gamma$ (`EtaTrivial.Feta`).

Theorem 7.5 (Located crossings lie on the line; `crossing_is_zero_on_line`). *If the eta-regularized limit of Theorem 5.3 vanishes at ordinate y , then $\zeta(\frac{1}{2} + iy) = 0$ and $\operatorname{Re}(\frac{1}{2} + iy) = \frac{1}{2}$.*

8 Faithfulness for every Dirichlet L -function and Tate completion

A single carrier serves all χ ; the character weight $\chi(n)$ on each phasor is the only dial.

Theorem 8.1 (One carrier, all characters; `faithful_all_L`). *For every Dirichlet character χ :*

- (i) *for $\operatorname{Re} s > 1$, $\sum_{n < N} \chi(n) n^{-s} \rightarrow L(s, \chi)$; and*
- (ii) *on the critical line $\operatorname{Re} s = \frac{1}{2}$, the eta-twisted closed carrier `etaTwistClosed`(χ, s) vanishes if and only if $L(s, \chi) = 0$ (`etaTwistClosed_eq_zero_iff_critical`).*

Definition 8.2 (Completed carrier; `Tate.completedCarrier`). *For a Dirichlet character χ , set $\Lambda_\chi^c(y) = \Lambda(\chi, \frac{1}{2} + iy)$ using the archimedean Γ -completion $\Lambda(\chi, s) = (\text{gamma factor}) \cdot L(s, \chi)$.*

Theorem 8.3 (Completion and functional equation; `faithful_all_L_completed, completedCarrier_eq_zero_iff_completed_functional_equation`). *For every primitive χ modulo q :*

- (i) *the completion is zero-free off the zeros of L on the line: $\Lambda_\chi^c(y) = 0 \iff L(\frac{1}{2} + iy, \chi) = 0$ (the Γ -factor never vanishes for $\operatorname{Re} s > 0$); and*
- (ii) *Tate's functional equation [10] holds: $\Lambda(\chi, 1-s) = q^{s-1/2} W(\chi) \Lambda(\chi^{-1}, s)$, where $W(\chi) = \text{rootNumber}(\chi)$.*

9 The double-ended carrier and conjugacy

The carrier is chiral: the right strand winds with spin $e^{-iy \log n}$, and its mirror, the left strand, winds with $e^{+iy \log n}$.

Theorem 9.1 (Conjugate chiralities; `both_helices_conjugate`). *For every $N \in \mathbb{N}$ and $y \in \mathbb{R}$,*

$$\sum_{n < N} (-1)^{n+1} n^{-(1/2-iy)} = \overline{\sum_{n < N} (-1)^{n+1} n^{-(1/2+iy)}}.$$

Proof sketch. Termwise: $\overline{(-1)^{n+1}} = (-1)^{n+1}$ and $\overline{n^{-(1/2+iy)}} = n^{-(1/2-iy)}$ for the positive real base n (so $\arg n \neq \pi$), by `Complex.cpow_conj` and $\bar{n} = n$. $\square$

Corollary 9.2 (Equal modulus, identical crossing heights). *The two strands have equal modulus at every N , so their limits vanish at the same ordinates y .*

Theorem 9.3 (Completed sides agree; `both_sides_cross_together`). *For every $y \in \mathbb{R}$, $\Lambda(\frac{1}{2} - iy) = \Lambda(\frac{1}{2} + iy)$.*

Proof sketch. $\frac{1}{2} - iy = 1 - (\frac{1}{2} + iy)$, so this is $\Lambda(1 - s) = \Lambda(s)$ (`completedRiemannZeta_one_sub`) on the line. $\square$

10 The Frobenius similitude and the determinant identity

The map $n \mapsto pn$ is the Frobenius self-similarity of the carrier. By Theorem 3.2 it acts on the carrier point $\text{helixPt}(n) = \sqrt{n} \text{wind}(n)$ as

$$\text{helixPt}(pn) = \sqrt{pn} \text{wind}(pn) = \underbrace{\sqrt{p}}_{\text{radial}} \underbrace{\text{wind}(p)}_{\text{rotation}} \text{helixPt}(n),$$

a similitude: radial scaling $\sqrt{p}$ (the area-law factor of Theorem 2.6) times a rotation by the unit-modulus winding $\text{wind}(p)$. Under the bridge (Theorem 3.4) the rotation eigenphase at ordinate y is the spin $z = \text{spin}(y, n) = e^{-iy \log n} \in \mathbb{T}$. The two chiralities (Theorem 9.1) carry z and $\bar{z}$.

Theorem 10.1 (Conjugate-eigenstate determinant; `frobenius_conjugate_det_one`). *For every $y \in \mathbb{R}$ and $n \in \mathbb{N}$, with $z = \text{spin}(y, n) = e^{-iy \log n}$,*

$$\det \begin{pmatrix} z & 0 \\ 0 & \bar{z} \end{pmatrix} = z \bar{z} = |z|^2 = 1.$$

Proof. The 2×2 determinant is $z \cdot \bar{z} - 0 \cdot 0 = z \bar{z}$. Since $\bar{z} = \overline{e^{-iy \log n}} = e^{+iy \log n} = e^{+iy \log n}$ and $-iy \log n + +iy \log n = 0$, we have $z \bar{z} = e^{-iy \log n} e^{+iy \log n} = e^0 = 1$. $\square$

Remark 10.2. *The transverse block $\text{diag}(z, \bar{z})$ assembled from the two chiral eigenphases at a conjugate crossing has determinant 1; the radial scaling $\sqrt{p}$ of the similitude is carried separately and is not part of this block. This is the terminal statement of the construction.*

11 Formalization

Every definition and theorem above is formalized in Lean 4 [12] against Mathlib [11], in the package `RequestProject` (files `ClosedForm.lean`, `HelixLogFreeFTA.lean`, `HelixCollapseReality.lean`, `LFunctionPhasor.lean`, `GeometricProjectionHolds.lean`, `Faithfulness.lean`). Each named theorem compiles with no `sorry`; the kernel axiom footprint reported by `#print axioms` (equivalently `lean.verify`) is

$$\{ \text{propext}, \text{Classical.choice}, \text{Quot.sound} \},$$

the three standard Mathlib axioms, with no construction-specific axiom. The correspondence between the statements here and the Lean declarations is given by the identifiers in parentheses throughout.

The capstone identities — the strip extension (Theorems 5.1, 5.2, 5.3), the conjugacy of the two chiralities (Theorem 9.1), and the determinant identity (Theorem 10.1) — are collected in `Faithfulness.lean`; the geometry (Section 2) and the log-free winding (Section 3) live in `ClosedForm.lean` and `HelixLogFreeFTA.lean`.

Remark 11.1 (Scope). *This paper presents a geometric tool: the double-ended chiral helix carrier and its phasor fiber, on which the vanishing points coincide with the zeros of ζ and the tested Dirichlet L -functions, verified numerically for the leading zeros. Its mathematical content is exactly the statements proved above — the carrier, the accumulation to $L(s, \chi)$ on $\operatorname{Re} s > 0$, the explicit-formula reading of the zeros as poles, the reality of the completed object on the critical line, the conjugacy of the two chiralities, and the determinant identity of Theorem 10.1.*

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